CMU-Africa MSIT + Mastercard Foundation Scholars: Application Draft

Complete draft for your review. Deadline: early round 15 Dec 2026 (submit by 12 Dec)

How to use this file:

1. Open the CMU-Africa applicant portal in your browser.
2. Copy each **exact** portal question above the matching draft below.
3. Rewrite each draft in your own words, keeping only facts you can prove.
4. Fill every `[⏳ …]`. Delete any sentence you cannot support.

Evidence:

- CMU-Africa facts are **E2**: found by search on africa.engineering.cmu.edu; the site fails its security check from this machine. Confirm them in the portal.
- Mastercard Foundation facts are **E1**.

1. Three separate decisions

	1. Admission (MSIT)	2. Need-based financial aid	3. Mastercard Foundation Scholars
Who decides	CMU-Africa admissions	CMU-Africa financial aid	CMU-Africa with the Mastercard Foundation
Based on	Programming ability, English, transcript, essays + video essay, CV, 3 recommenders, technical test	Financial need (form after you apply)	Academic talent, giving back, leadership; people facing the highest barriers
Your weak point	GPA 2.62 (most admits have ≥ 3.0)	Must document need honestly	“Academic talent” must be shown beyond the CGPA
Your strongest evidence	Final-year 3.20; A− in Advanced Programming, Compiler Design, Complexity Theory, Final Project I and II; fintech infrastructure work; public code	Real income vs US$16,000 tuition + living costs	Service-oriented work in payments; real teaching/mentoring; credible return plan

2. What the portal asks for (E2, confirm in the portal)

Item	Rule	Your file
Online application	Personal and education data	Use block 3 below
Written essays	Several questions answered in the portal	Section 5 drafts
Video essay	One question, recorded in the portal; **cannot be edited or re-recorded**; Chrome or Firefox	Section 10: practise first
CV	**PDF, at most 2 pages** (pages beyond 2 are not considered)	`06_FINAL_CV/CV_CMU-Africa_2pages.docx`
3 recommenders	Contact details only; CMU contacts them (no letters at submission)	Project supervisor, A− lecturer, SantimPay manager
Transcripts	Unofficial, with grading key, all years in one PDF; official transcripts by 30 Jun 2027 if admitted	DTU transcript scan
English	DET/TOEFL/IELTS **optional** if you apply by 15 Dec; DET fee covered for first-time early applicants	Do not book a test for CMU

Item	Rule	Your file
Technical test	After the application is complete: analytical, quantitative, programming logic; fee covered by CMU	Section 11
Fee	No application fee	–

3. Academic history (copy into the form)

> Debre Tabor University, Faculty of Technology, Ethiopia: BSc in Computer Science (Extension programme), academic years 2017/18 – 2021/22, awarded 14 July 2022. Cumulative GPA **2.62 / 4.00** (as printed on the transcript). Language of instruction: English.

GPA rule: enter **2.62 / 4.00 as printed**; do not recalculate or round. The final-year figure (**3.20**) is your own calculation from the transcript. Use it only in essays and say so.

Course highlights:

- Advanced Programming A−
- Compiler Design A−
- Complexity Theory A−
- Final Project I A− and Final Project II A−
- Selected Topics in CS A
- Wireless and Mobile Computing B+
- Human–Computer Interaction B+
- Computer Security B
- Introduction to Distributed Systems B−

4. Professional CV

Use `06_FINAL_CV/CV_CMU-Africa_2pages.docx`. It must stay at two pages; export it to PDF. Job titles, dates and duties stay as `[⏳]` until the HR letters arrive.

5. Essay drafts (about 250–300 words each)

Essay A: Why the MSIT, why CMU-Africa, and how you will use it

> **Portal question:** [paste]

At SantimPay in Addis Ababa I work on the infrastructure behind digital payments: card and QR payments at shops, school-fee payments, and payment services that partner banks offer their customers [⏳ your exact role from the HR letter]. When a server, a deployment or a network link fails, a merchant cannot take a payment. I have learned to keep these systems running. I have also learned how much I do not yet understand about designing them.

Most of my work is operational: I [⏳ two verified duties, e.g. automate server configuration with Ansible; maintain the deployment pipeline for … services]. In [⏳ month/year], [⏳ one real incident or problem: what failed, how you found the cause, what changed afterwards]. I could fix the immediate problem, but I lacked the depth in distributed systems, security engineering and architecture to redesign the system so that failures of that kind stop happening.

My undergraduate record began weakly and improved: my final year was my strongest (GPA 3.20 by my calculation from the transcript), with A− grades in Advanced Programming, Compiler Design, Complexity Theory and both parts of my final-year project ([⏳ title]).

The MSIT combines Carnegie Mellon's engineering rigour with problems specific to African markets. [⏳ Name one or two MSIT courses or concentrations from the official curriculum and say why each matters to your work.] I also want to learn alongside engineers from across the continent who face the same constraints: unreliable connectivity, fast-growing transaction volumes, and the need for security that regulators and customers trust.

After the MSIT I intend to design, not only maintain, payment infrastructure in Ethiopia: [⏳ a concrete role, e.g. leading reliability and security engineering for a payments platform].

Essay B: Professional journey and leadership

> **Portal question:** [paste]

I graduated in Computer Science from Debre Tabor University in July 2022 and joined SantimPay in [⏳ month year per HR letter] as [⏳ title]. [⏳ Later titles with dates exactly as in the HR letter.] [⏳ One or two sentences on Kifiya: dates and role exactly as in Kifiya's letter.]

One leadership example (only a real one; delete this part if none applies):

- **Situation:** [⏳ e.g. rollout of POS terminals to N merchants, or a recurring outage]
- **My mandate:** [⏳ what you were responsible for; how many people you coordinated]
- **What I decided and did:** [⏳ two or three concrete actions]
- **Result:** [⏳ what changed; a number only if you can prove it]
- **What I learned:** [⏳ one sentence]

These experiences taught me that reliable technology depends as much on coordination and clear communication as on code.

Essay C: Contribution to Africa, giving back, and your community

> **Portal question:** [paste]

In Ethiopia many merchants and families are only now moving from cash to digital payments. When a payment system fails, they lose a sale, and sometimes their trust in digital money. The infrastructure behind those payments is the part of the economy I know best.

My goal is to help build payment infrastructure in Ethiopia that is secure and dependable enough for that trust to grow. Within five years of the MSIT, I aim to [⏳ realistic goal].

Giving back, for me, means sharing what I learn. In 2022 I worked as a co-trainer with Dereja, supporting trainees on the learning platform and preparing weekly progress reports [⏳ confirm dates, number of trainees and whether you have a certificate]. After the MSIT I will [⏳ one real, modest commitment, e.g. run a free systems-reliability workshop each semester for Computer Science students at Debre Tabor University; mentor junior engineers at my workplace].

Optional essay D: Academic record (only if a question invites context)

My cumulative GPA of 2.62/4.00 does not show the direction of my studies. I completed my BSc through Debre Tabor University's extension programme over five academic years. My first years were weak: I failed my first attempts at Probability and Statistics and at Discrete Mathematics, and later passed both. As I moved into core computer-science subjects my results rose. My final-year GPA was 3.20 (my calculation from the transcript), with A− grades in Compiler Design, Advanced Programming, Complexity Theory and both parts of my final-year project. I know mathematics is my weaker foundation. I am addressing it by [⏳ a real step only, e.g. completing a discrete-mathematics course before enrolment].

6. Professional achievements: evidence bank

Use one or two of these, and only once you can prove them:

Achievement area	What to write (fill only with real facts)	Proof
Production reliability	[⏳ services or servers you operate; uptime or incidents handled]	HR letter, manager
Automation	[⏳ what you automated with Ansible; time saved, if measured]	Manager, repository history
Delivery pipeline	[⏳ pipelines you built or maintained; deployment frequency]	Manager
Merchant/POS operations	[⏳ terminals or merchants, your role]	Records you may cite
Security operations (Kifiya)	[⏳ controls you configured yourself]	Kifiya letter

Achievement area	What to write (fill only with real facts)	Proof
Certification	Certified Kubernetes Administrator (2022, **expired 2025**). A hands-on exam; list it as expired, with its ID	Certificate ID [⏳]

7. Technical portfolio (for the CV and interview)

- **Final-year project:** [⏳ title], A− and A−. One-paragraph description once you have the report.
- **Public demo 1, CI/CD + configuration automation:** your own GitHub repository (spec in `09_AISS/AISS_Portfolio.md`, project P5). Put the link on the CV.
- **Public demo 2, monitoring and alerting:** spec in the same file (P6).

8. Financial need statement (truthful structure)

The Mastercard Foundation verifies information. Use exact figures you can document.

Item	Amount	Source
Your monthly net salary (ETB)	[⏳]	Payslip
Other household income (ETB/month)	[⏳]	–
People who depend on you	[⏳]	–
Savings and assets	[⏳]	–
Tuition for African students, 2026–27	**US$16,000**	E2, CMU-Africa tuition page
Living costs in Kigali	[⏳ from CMU-Africa's published estimate]	E2
Travel and visa	[⏳]	–

> **Draft paragraph:** I earn [⏳ amount] per month, which supports [⏳ who]. My savings are [⏳]. Tuition of US$16,000 plus living costs in Kigali are [⏳ N] times my annual income. Without the Mastercard Foundation Scholarship I cannot enrol. I have also applied to [list other scholarships]; none is secured.

Do not understate income, hide employment or describe hardship you have not experienced.

9. Leadership and community impact (Mastercard Foundation focus)

What the Foundation looks for (E1 target group; E2 criteria):

- "highly talented, service-oriented young people who face the highest social and economic barriers to opportunity"
- academic talent
- commitment to giving back
- leadership potential

Theme	How to show it truthfully	Gap to close before 12 Dec
Talent	Final-year trend; A− courses; Kubernetes exam passed (2022); public code	Build demo 1
Service orientation	Your work keeps merchants and families able to pay	One real merchant- or user-impact example
Leadership	One STAR example where you led, coordinated or trained people	Confirm the example with your manager
Giving back	Dereja co-training (confirm); a concrete plan	Get the Dereja certificate or a contact

10. Video essay (cannot be re-recorded)

Practise this routine five times before the real recording:

- 60–90 seconds
- one story
- one lesson
- one plan

> I keep payment systems running for merchants in Ethiopia [⏳ verified]. The moment I knew I needed this degree was [⏳ real incident, one sentence]. I want CMU-Africa to give me the design skills to prevent such failures, not just fix them, and I will share them back home by [⏳ real commitment].

Setup:

- Chrome or Firefox
- quiet room, light facing you
- camera at eye level
- no script on screen

11. Technical test (after the application is complete)

Content (E2): analytical and quantitative skills and programming logic; the fee is covered.

Plan: 2 weeks, 1 hour a day.

- **Days 1–4:** arrays, strings, loops, recursion and basic complexity, in Python.
- **Days 5–8:** logic puzzles, number sequences, percentages and rates.
- **Days 9–12:** timed mixed sets of 20 questions.
- **Days 13–14:** rest and review your mistakes.

12. Interview (if invited)

Question	Outline
Walk me through your background	Degree → final-year improvement → role → one result
Why do you need this degree now?	The design gap you hit (incident story)
Your GPA is below most admits	Trend, A− courses, Kubernetes exam, production responsibility, public code
Lead / failure / conflict example	STAR stories from section 9
How will you give back?	The concrete commitment from essay C
Explain a tool on your CV	Only tools you can explain in depth
How would you secure a payment API?	Authentication, TLS, rate limiting, logging, least privilege

13. Before you press Submit

- [ ] Passport name everywhere
- [ ] CV ≤ 2 pages, as a PDF; dates match the HR letters
- [ ] GPA entered as 2.62/4.00
- [ ] Every essay in your words and within the word limit
- [ ] Three recommenders informed (CMU will contact them)
- [ ] Video practised (no re-record)
- [ ] Submitted by **12 Dec** (early deadline 15 Dec)
- [ ] Afterwards: financial aid form; technical test when invited